ELEC70142

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2026

Digital VLSI System-on-Chip (SoC) Design

Monday, 11 May 2026, 10:00 AM

Time allowed: 2:00 hours

There are FOUR questions on the paper.

Answer ALL questions.

Any special instructions for invigilators and information for Candidates are on page 1.

Examiners responsible First Marker(s): P.Y.K. Cheung

Second Marker(s): Christos Bouganis

Information for Invigilators:

Students may bring ONE A4 sheet of handwritten notes into the examination.

Information for Candidates:

Students may need red, green, blue, yellow and black coloured pens.

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1. *Figure Q1* shows the layout of a digital circuit using an n-well process. It consists of 14 transistors, labelled in yellow.

   a) What logic functions are performed by transistors T11, T12, T13 and T14?

   [5 marks]

   b) Extract the transistor level schematic circuit for T1 to T4. What logic function does this circuit perform? (Sizes of transistors not required.)

   [5 marks]

   c) Transistors T5 to T10 form a compound gate. Extract the transistor level schematic circuit for T5 to T10. (Sizes of transistors not required.)

   [10 marks]

   d) Based on the answers above, derive the logic equation for OUT3 in terms of input signals A and B.

   [5 marks]

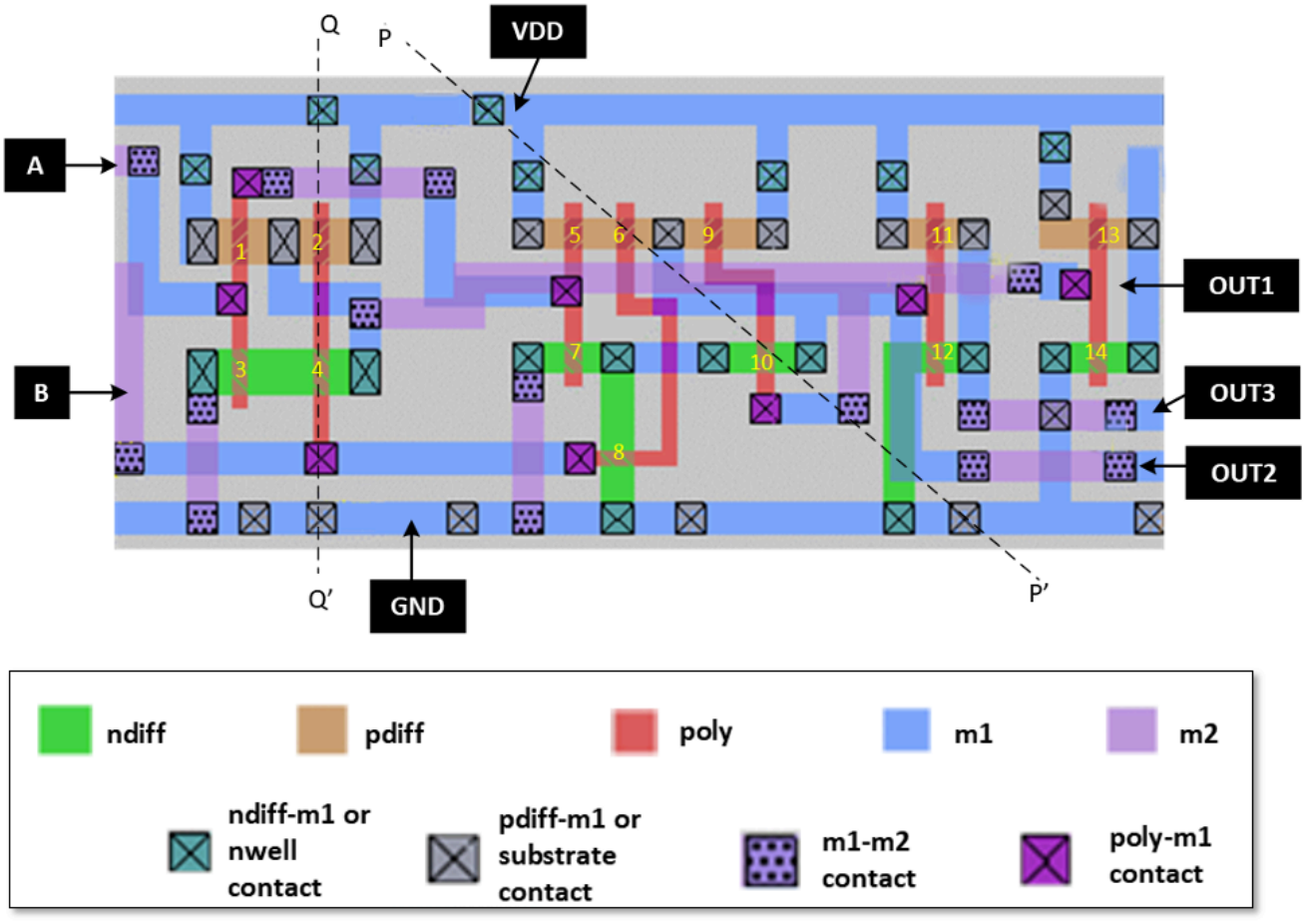
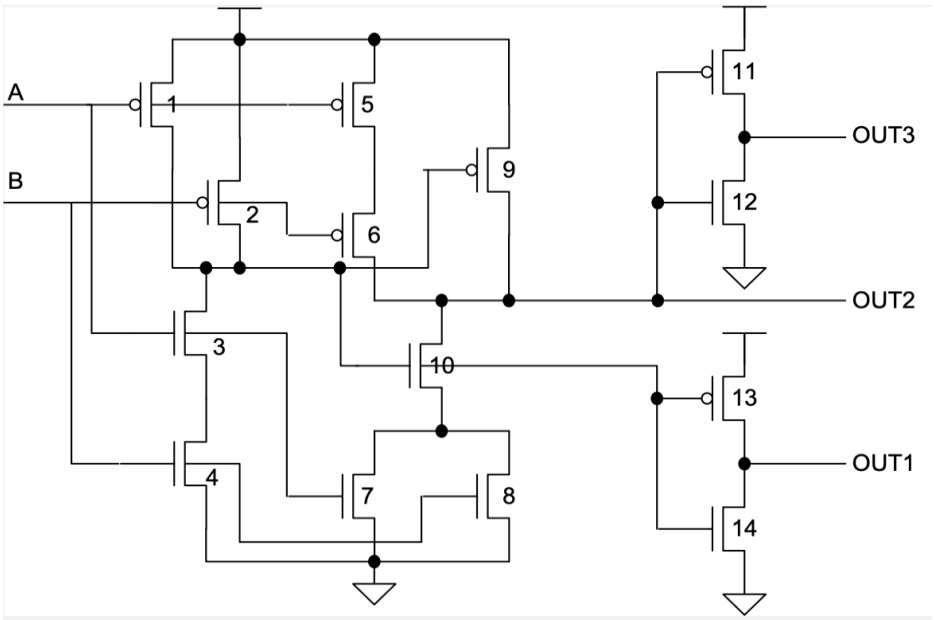


Figure Q1 Layout of a full-custom cell (Q1 & Q2)

Solution to Q1

This question tests student's ability to understand a full custom layout of digital logic gates. The question is deliberated divided into separate sections to help students to progress towards the final answer. The transistor level schematic (with sizes, but not asked) of the entire circuit is:



a) T11 to T14 form two inverter circuits.

[5 marks]

b) T1 to T4 is a two input NAND gate with inputs A and B. See above for the schematic circuit.

[5 marks]

c) T5 to T10 is a compound gate that has three inputs: A, B and NAND(A,B). Schematic is as shown above.

[10 marks]

d) The Boolean equation of OUT3 is:

$$OUT3 = (A + B) \cdot \overline{(A \,\&\, B)}$$

[5 marks]

2. (a) For the layout shown in *Figure Q1* above, draw the vertical cross section along the line Q and Q′. Label your diagram indicating the n-well region, the different types and levels of doping (e.g. p^-, n^+ etc) and the different metallisation. You do not need to draw to scale.

[15 marks]

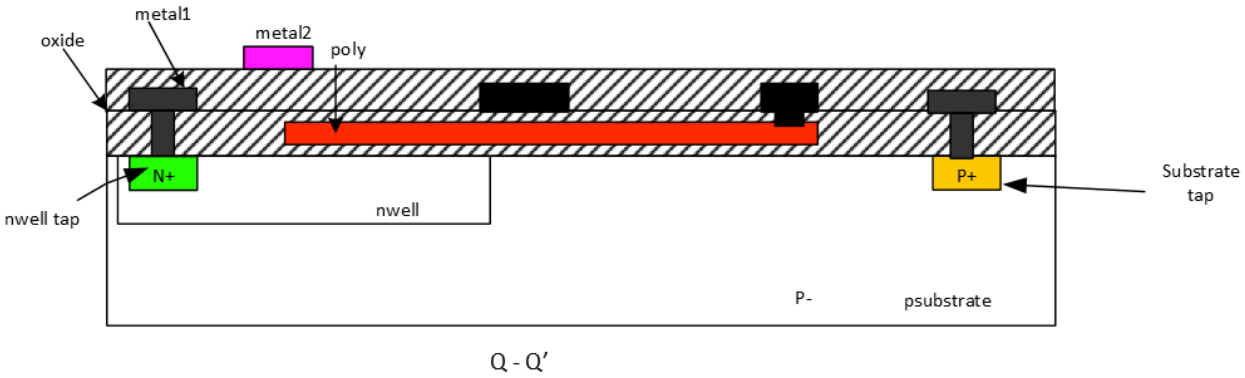
(b) Briefly explain the purpose of the nwell and substrate contacts.

[10 marks]

Solution to Q2

This question tests student's understanding of the fabrication process and the different regions in an integrated circuits that create the transistors.

(a)

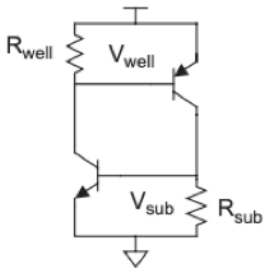


[15 marks]

(b)

Key points in answer:

- P+ diffusion of p-MOS transistor, N-well and P- substrate form a parasitic PNP transistor.
- N+ diffusion of n-MOS transistor, P- substrate and N-well form another parasitic NPN transistor.
- These two transistors form a thyristor circuit shown here where both transistors are not normally conducting.
- However, if Rwell and Rsub becomes large, transient current may momentarily turn one of these transistors ON, then the thyristor circuit will starts conducting, shorting the power rails. This is known as latch up.
- This is likely to eventually kill the integrated circuit.
- The N-well and substrate contacts provide low resistance path to ensure that the base-emitter junctions of the NPN, PNP transistors never develop a large enough voltage to turn on either transistor. This latch up is prevented.



[10 marks]

3. a) Explain briefly the advantages and limitations of the "stuck-at" model used in digital test.

[5 marks]

b) *Figure Q3(a)* shows the schematic circuit of a 2-input gate with a manufacturing defect manifested as a break in the signal track labelled “Open”. Explain why this fault cannot be represented as a stuck-at fault. Derive the faulty behaviour of this circuit as a Boolean equation or a truth-table.

[10 marks]

c) By applying the principle of path sensitisation to the circuit shown in *Figure 3(b)*, generate the test vectors for detecting: i) a stuck-at-0 fault at node F, ii) a stuck-at-1 fault at node G.

[10 marks]

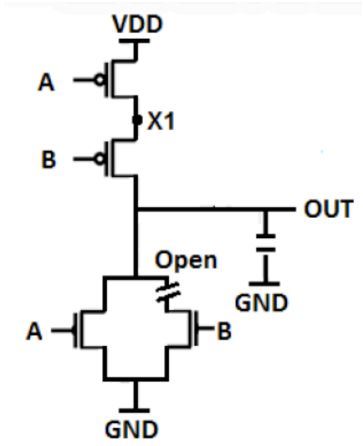


Figure Q3(a)

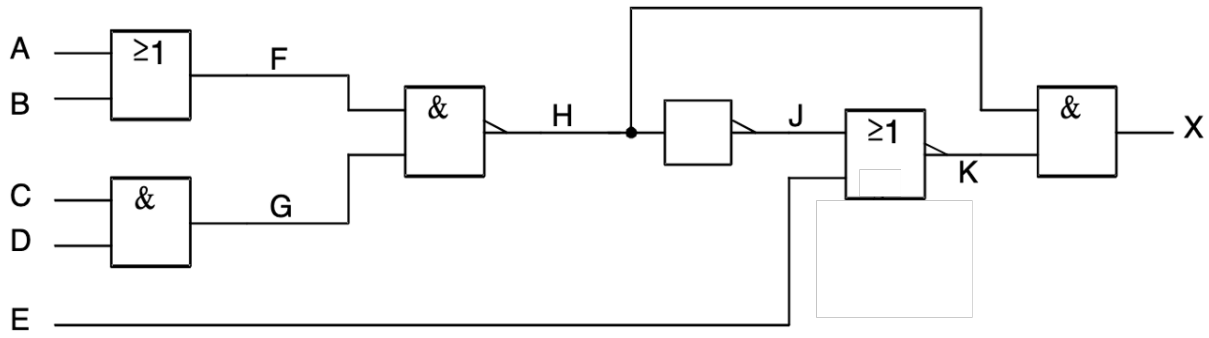


Figure Q3(b)

Solution to Q3

This question tests student's understanding of fault models, how such fault model is applied to actual digital circuits, limitations when applied to open-faults, and how the ATPG algorithm generate test vectors.

a)

Advantages:

- Can represent many physical faults such as shorts to Vdd & ground
- Easy to handle and simulate
- Many other faults can be mapped to stuck-at faults

Disadvantages:

- Cannot represent some bridging faults and some open-faults
- Cannot handle transient and delay faults

[5 marks]

b)

This is a 2-input NOR gate. The broken track fault cannot be represented by stuck-at model because it exhibits the behaviour of a sequential circuit. The faulty behaviour is:

$X = \overline{(P+Q)} + (P \bullet \overline{Q} \bullet X_{previous})$ or

A	B	OUT	OUT*
1	0	1	1
1	1	0	Z
1	0	0	0
1	1	0	0

[10 marks]

c)

Test vector to test Stuck-at-0 at node F:
{ABCDEX} = {1x1100}

Test vector to test stuck-at-1 at node G:
{ABCDEX} = {1x0x01}

[10 marks]

4. a) Explain the principle behind the operation of a pulse-triggered latch and its advantages and disadvantages.

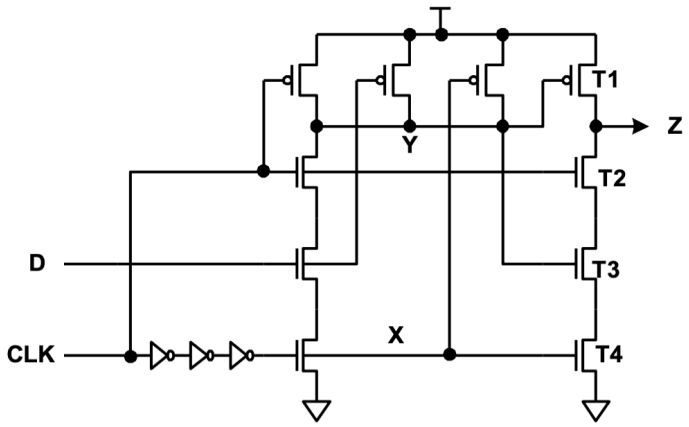
[7 marks]

b) *Figure Q4*(a) shows a synchronous circuit used in the AMD K-6 microprocessor. *Figure Q4(b)* shows the signals at `CLK` and `D`. Draw a timing diagram showing the signals `CLK`, `D`, `X`, `Y` and `Z`, and the state of transistors `T1`, `T2`, `T3` and `T4`. The timing diagram should be labelled with the state of the transistors {on, off} and with the following states for the signals:

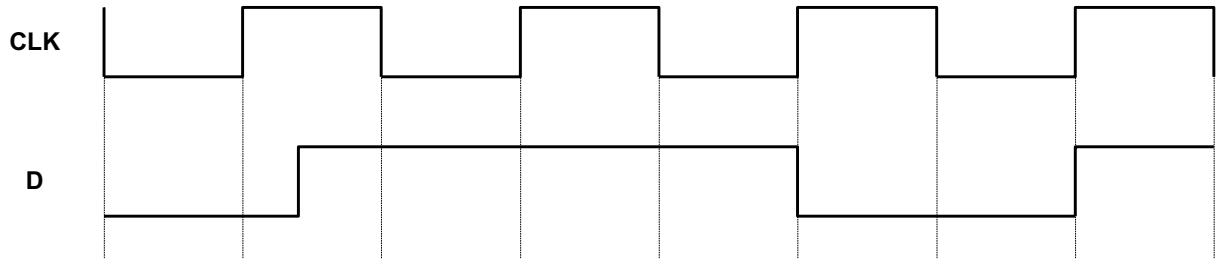
:

driven low (DL)	- a path to ground exists,
driven high (DH)	- a path to Vdd exists,
charged low (CL)	- high impedance with no stored charge,
charged high (CH)	- high impedance with stored charge at Vdd,
don't know (X)	- unknown level.

[18 marks]



Question Q4(a)



Question Q4(b)

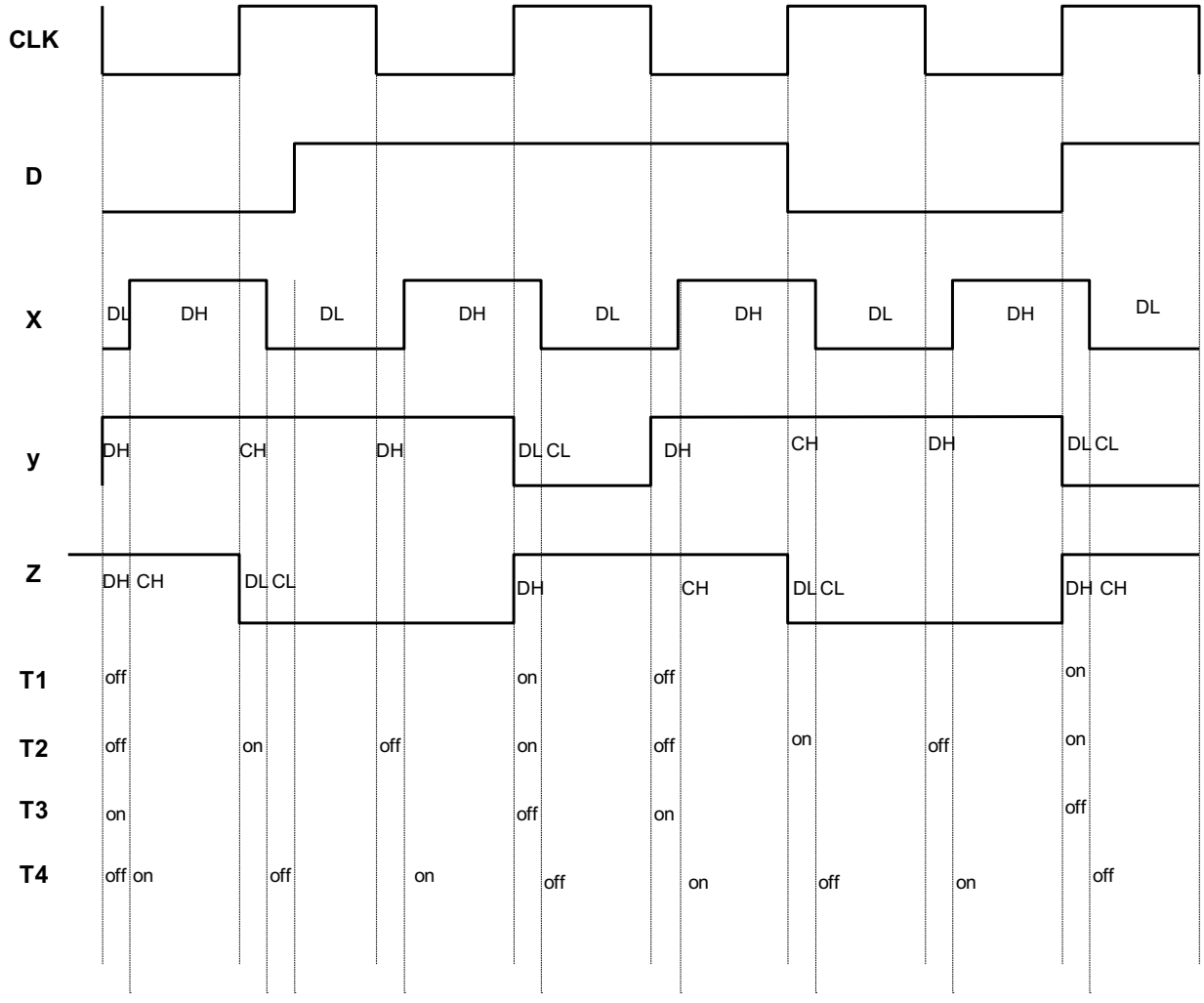
Solution to Q4

This question tests student's understanding of the following topics: clocked digital circuits, pulse triggered latch and flip-flop, dynamic circuit and state of digital signal beyond just 1's and 0's.

(a) A pulse-trigger latch/flip-flop consists of a pulse generator connected to the clock signal. A valid clock edge triggers a pulse in the latch/register which samples and stores the input data value. The advantage of the pulse-trigger latch is that it can yield negative setup time. The disadvantage is that the pulse width may not be that well controlled and the pulse generation circuit dissipates power no matter if the input data is changed or not.

[7 marks]

(b)



[18 marks]